

Sistemas Operativos

Algo de Sistemas Operativos,
Arquitecturas y Paralelismo

¿Qué es un Sistema Operativo?

- **Administra recursos**
- Arma una máquina virtual (visión y función)
- Comunicación con el usuario (visión y estímulos)
- Facilita el uso (herramientas, bibliotecas, etc.)

Funciones de un Sistema Operativo

Secuenciar las tareas (dispararlas)

Interpretar un lenguaje de control y comandos y estímulos

Administrar errores

Administrar las interrupciones

Scheduling

Proteger (administrar recursos)

Comunicación con el usuario

Características de un Sistema Operativo

- Concurrencia
- Uso compartido de recursos
- Almacenamiento a largo plazo
- Determinismo VS Indeterminismo
- Eficiente – Fiable
- Facilidad de corrección
- Tamaño Pequeño

Repaso Vocabulario

- Monoprocesador
Monoprogramación o Monotarea
(DOS – [trick TSR])

Multiprogramación o Multitarea
Batch

- Time-Sharing

Interactivo (Terminales Bobas o Inteligentes)

Repaso Vocabulario

Multiprogramación o Multitarea

UNIX (en todos sus sabores)

Linux – IRIX – AIX – BSD – Solaris – etc.

Windows (en casi todos sus sabores)

(Terminal Service)

zVM zOS

VMS

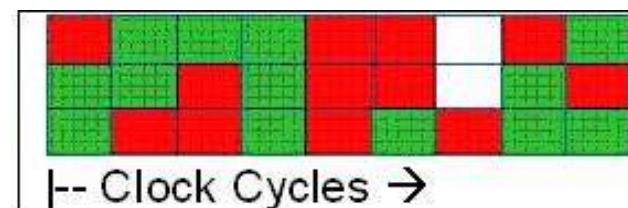
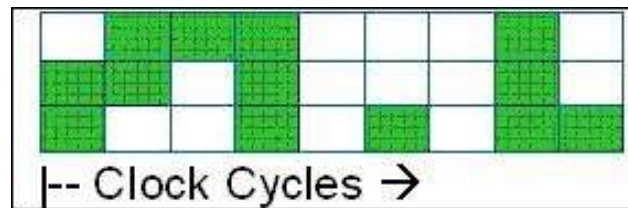
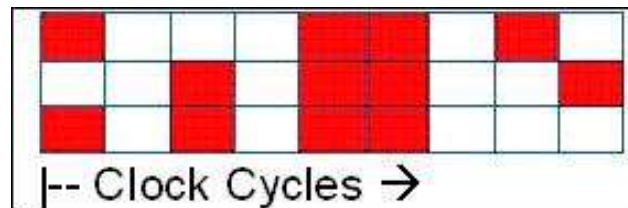
Repaso Vocabulario

- Multiprocesador
Varios Procesadores
Independientes o
Independientes con
Cores (Dual Quad Octo etc.)
→ Multiprocesamiento
SMP vs MP

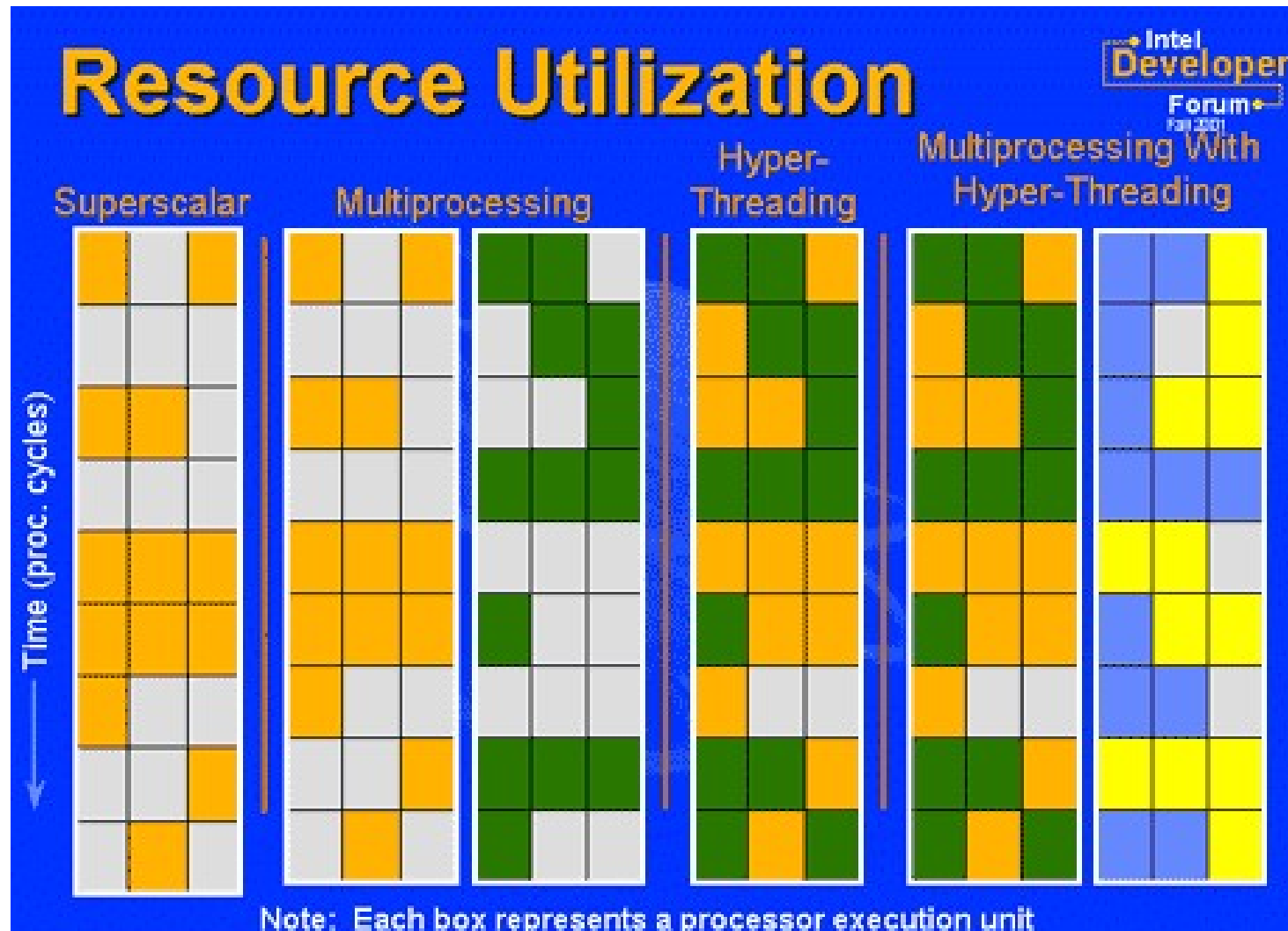
Repaso Vocabulario

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SMP vs MP
- * Hyper Threading ?

Hyper Threading



Hyper Threading y otros



Repaso Vocabulario

- Multiprocesador
Memoria Única
(Fuertemente Acoplado)

Memoria Distribuida
(Debilmente Acoplado) (en Red)

Sistemas Distribuidos o Multicomputadoras

Repaso Vocabulario

- Memorias

UMA

NUMA

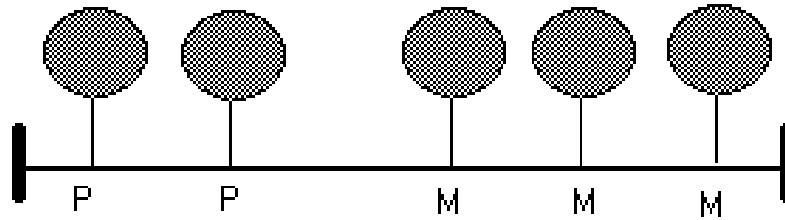
NORMA

COMA

cc-NUMA o SMP2

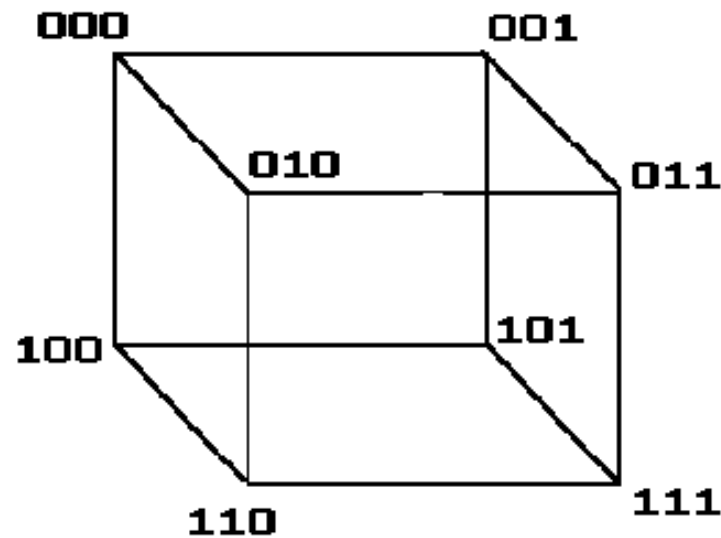
Repaso Vocabulario

- UMA



Repaso Vocabulario

- NUMA/NORMA/ccNUMA



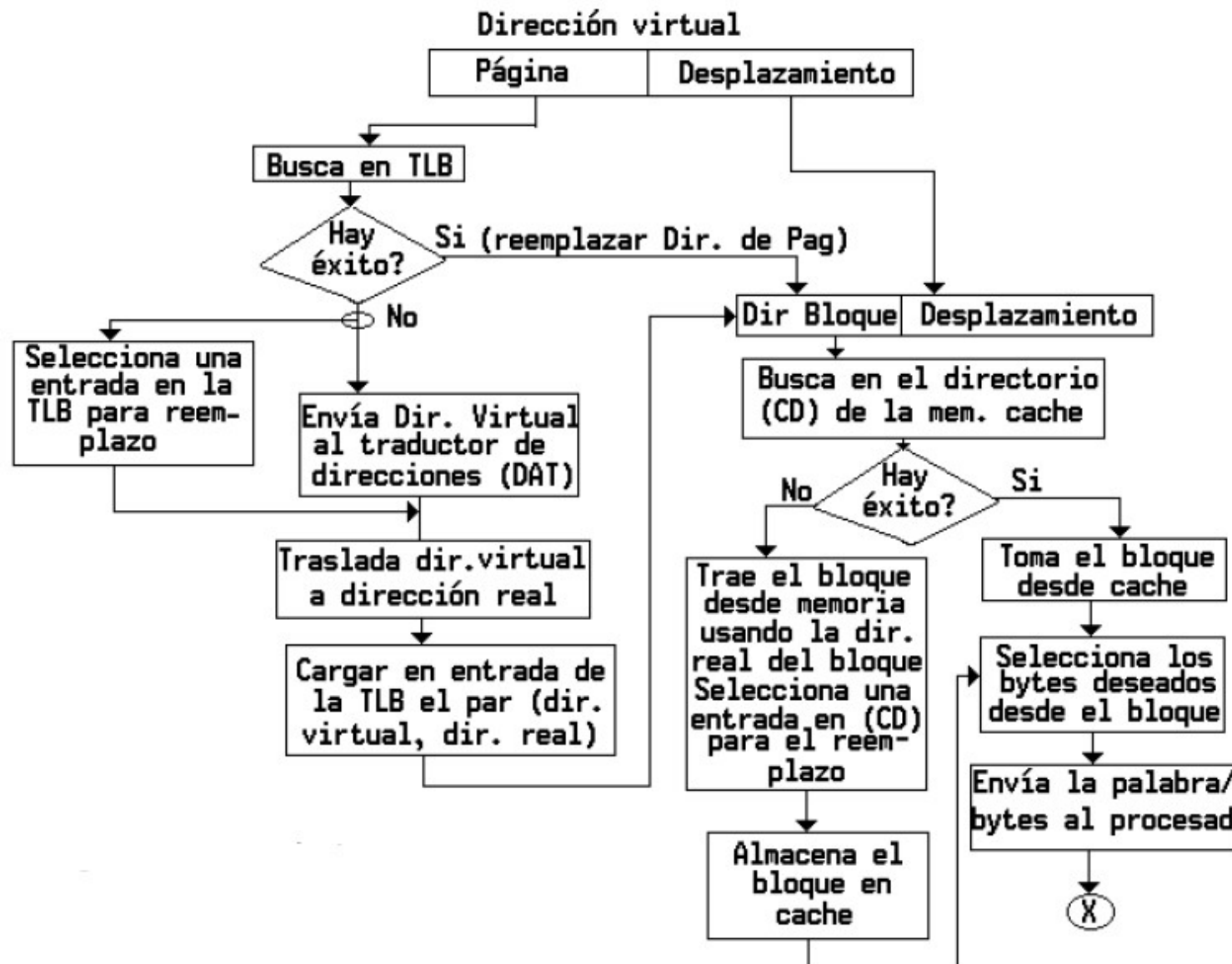
N-cubo de grado 3.

Repaso Vocabulario

- CACHE (bus compartido)
Snoopy - Write-through
Consistencia
00 Limpia
01 Compartida
10 Sucia por este procesador
11 Inválida en este procesador

¿Qué pasa sino se comparte bus?

Repaso Vocabulario: Memoria Cache (TLB)



Repaso Vocabulario

- Sistemas Distribuidos (sirven para ...)

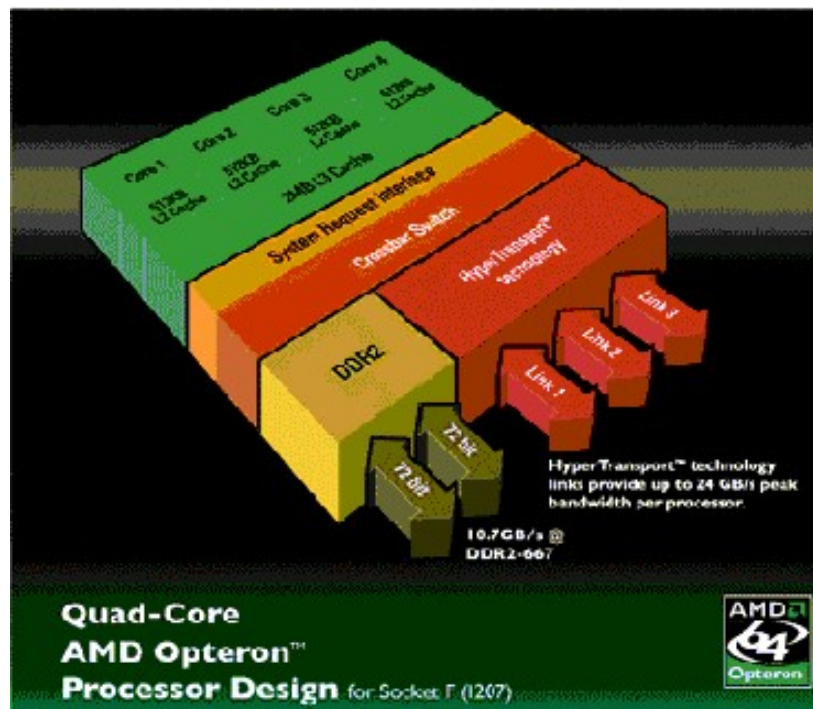
Cooperar (MPI, PVM, DSM, etc.)

Compartir (archivos, directorios, etc.)

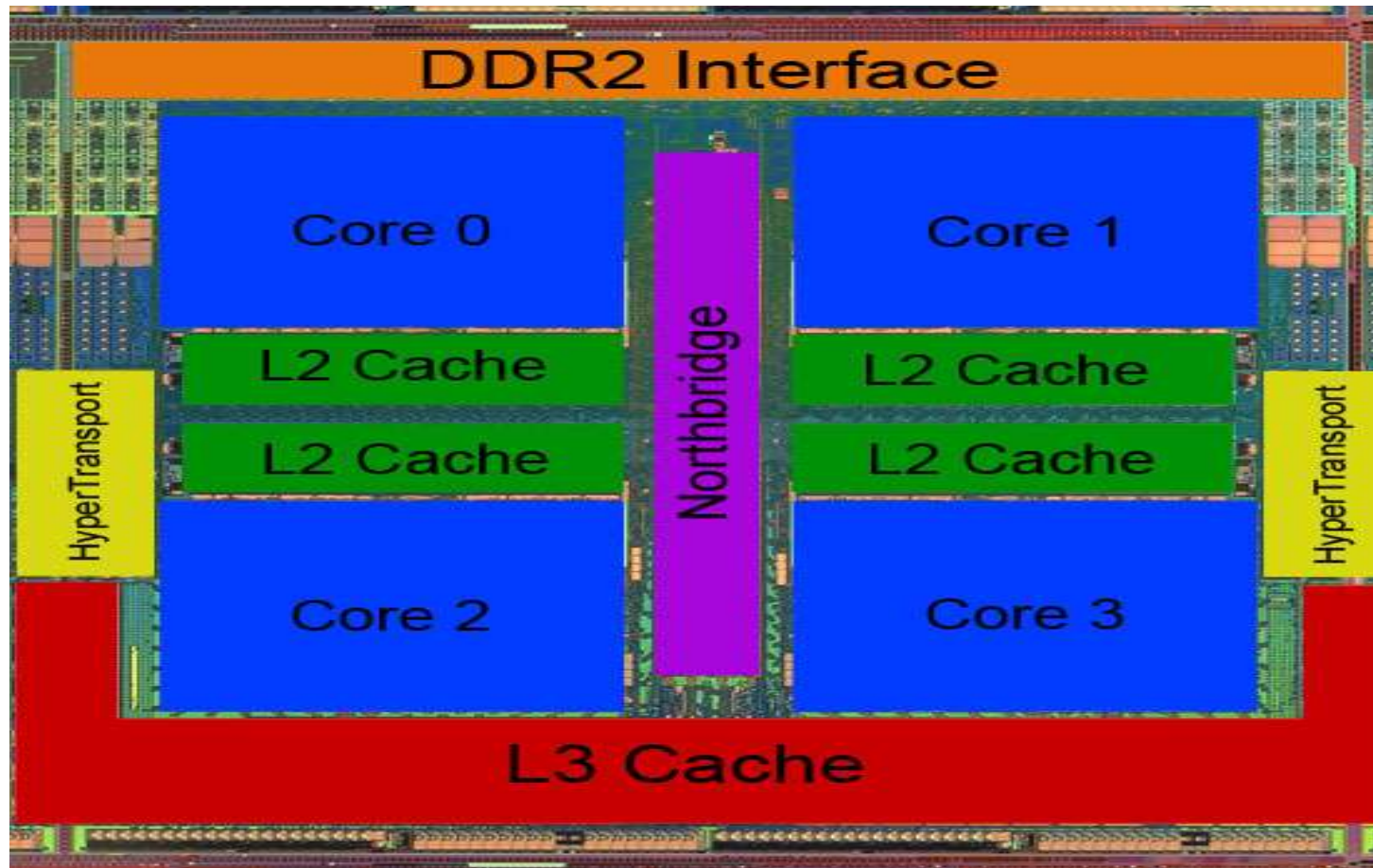
Servidores (Modelo Cliente/Servidor,
RPC)

Clusters → GRID → CLOUD

Cores



Cores



Cores

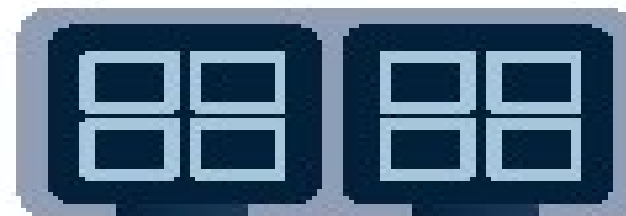
Quad-Core or Dual-Core

New 45nm

- 5400 Series
- 5200 Series

Up to 2x 1600 MHz

Intel®
ESB2
I/O
Bridge



Intel® Core™ Processor

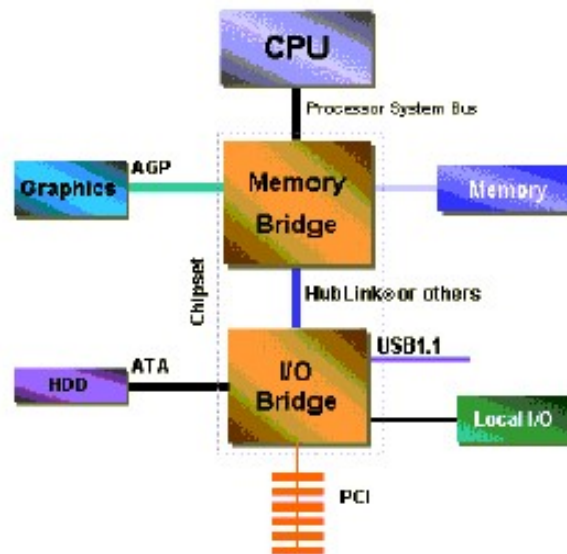
16 slots

128 GB max

Directly Buffered
Memory

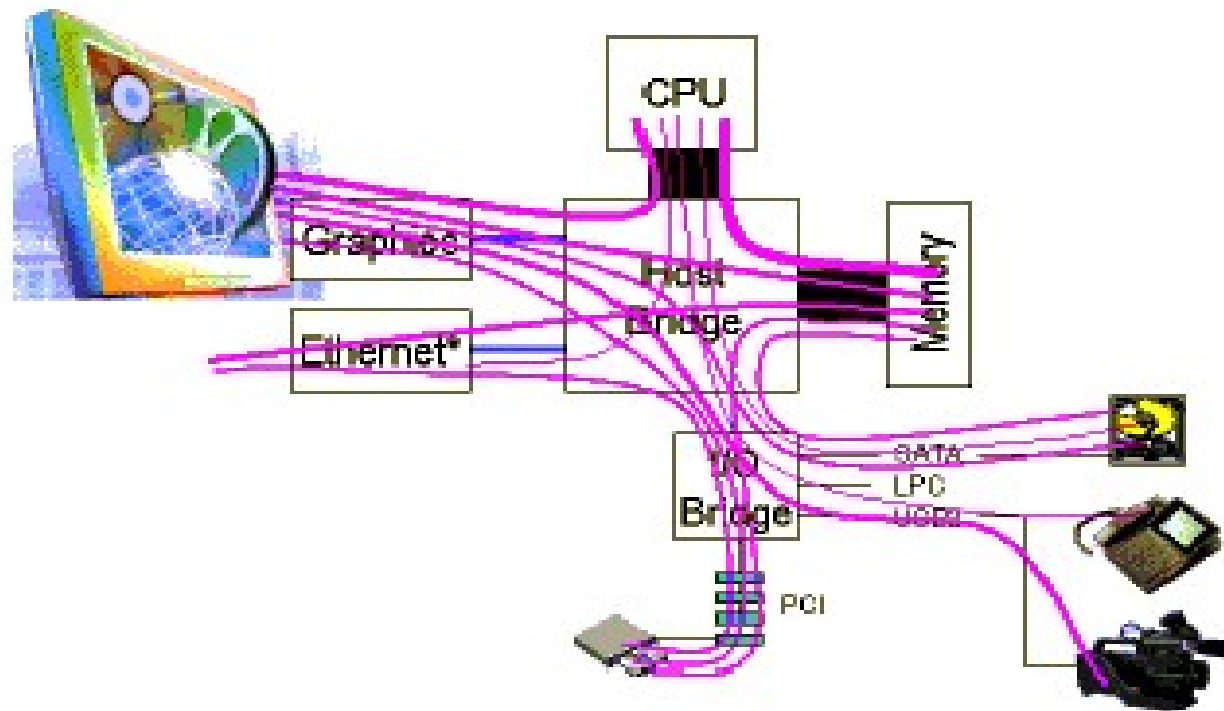
Configurable PCI Express® with 40 lanes
Dual x16 PCIe Gen 2

PCIs



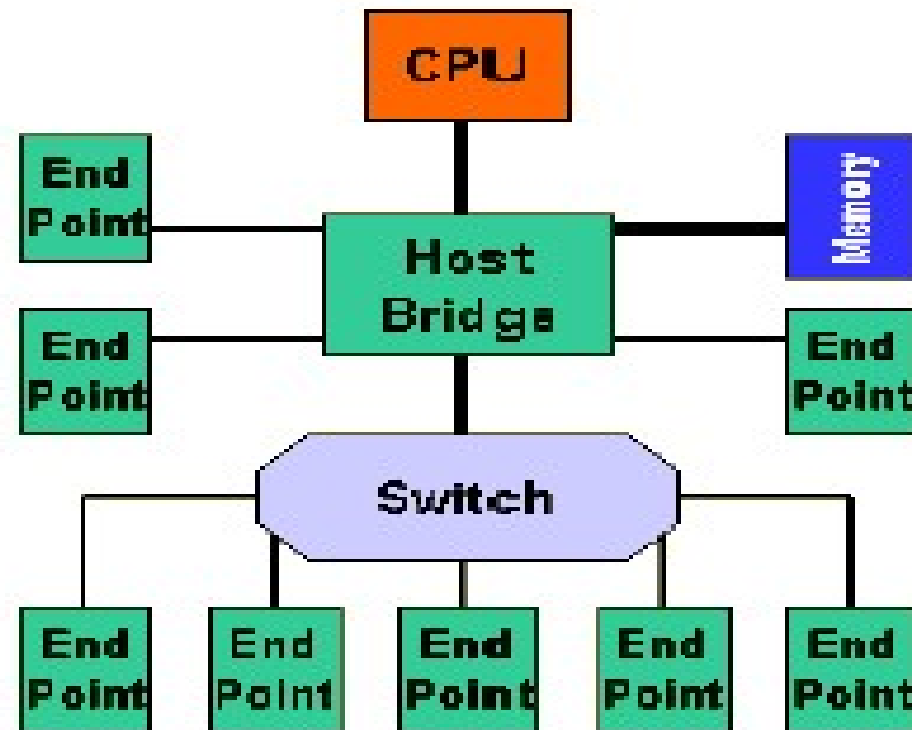
Today's PC has multiple local buses with different requirements

PCIs



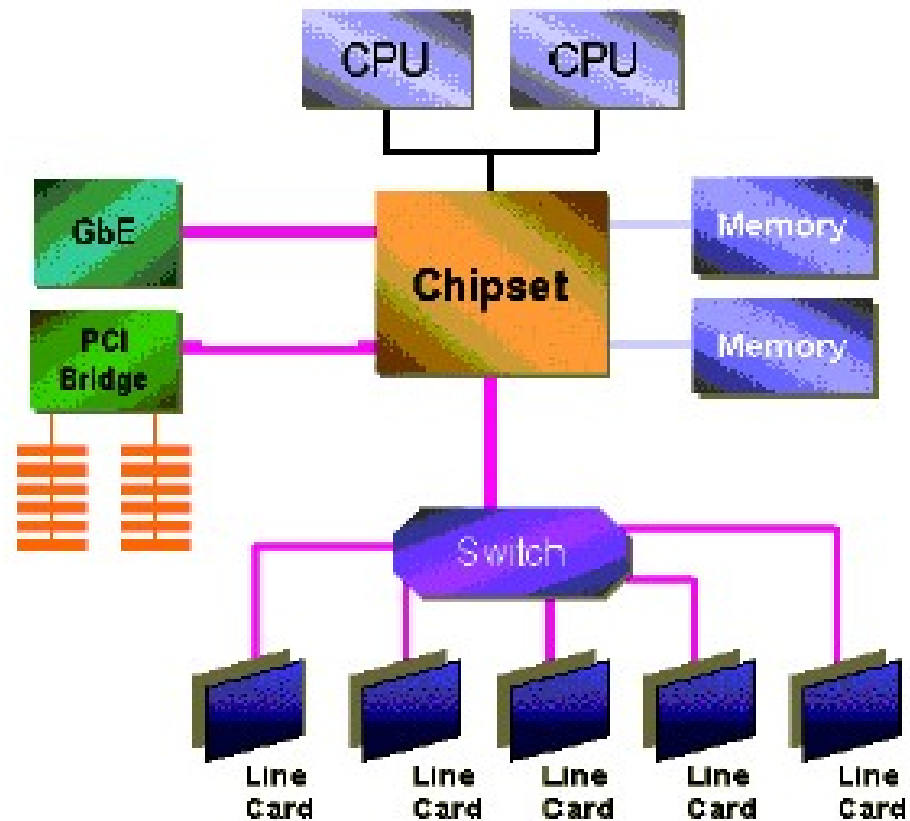
Multiple concurrent data transfers.

Mejora de Comunicación (Switch)



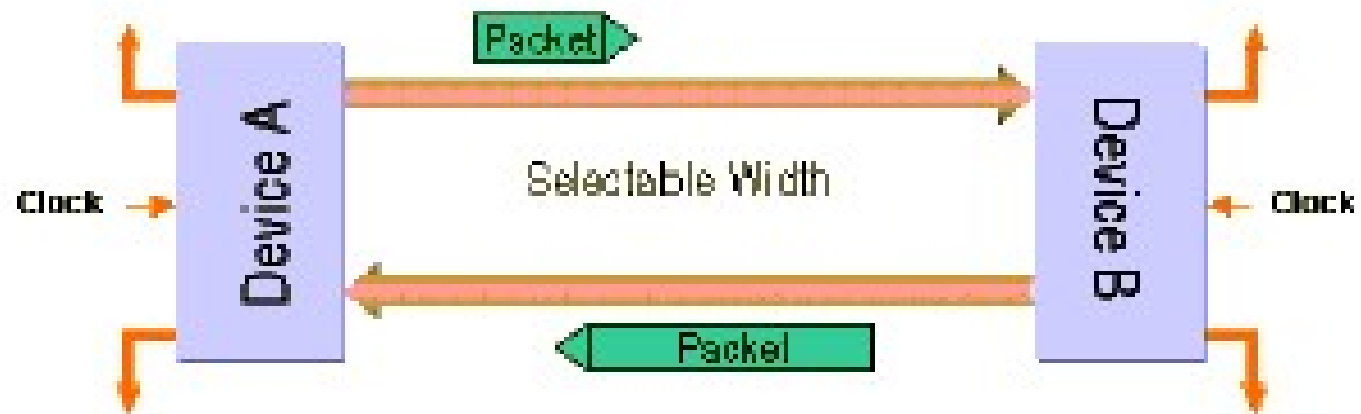
A switch is added to the system topology

PCI Express



PCI Express-based Networking Communications System

PCI Express



PCI Express link uses transmit and receive signal pairs

Usa más caminos

PCI

PCI

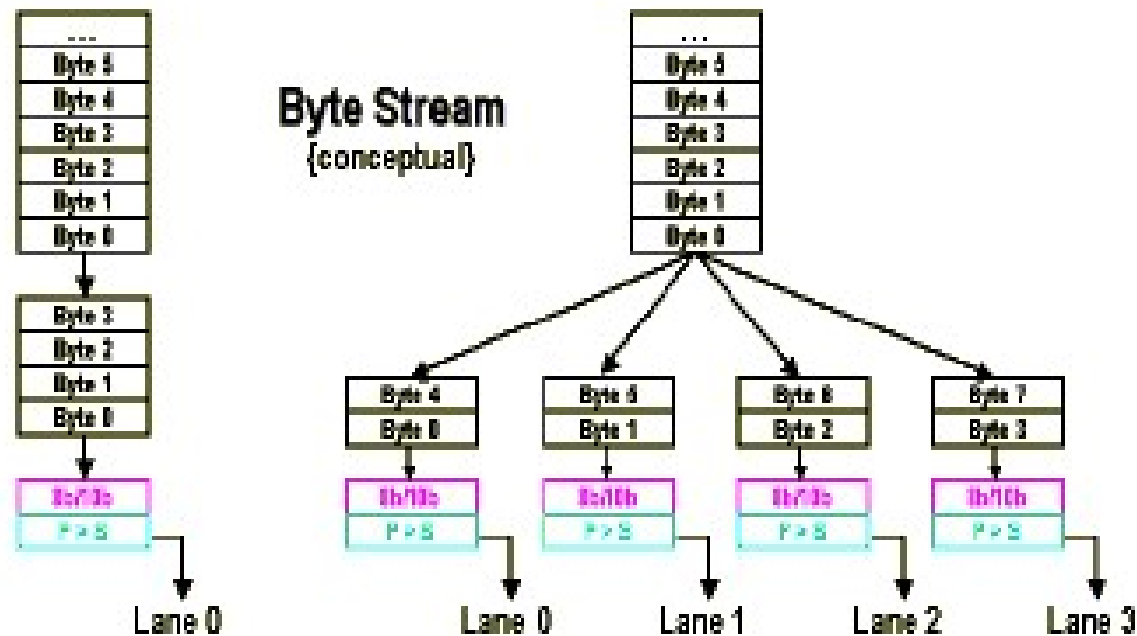
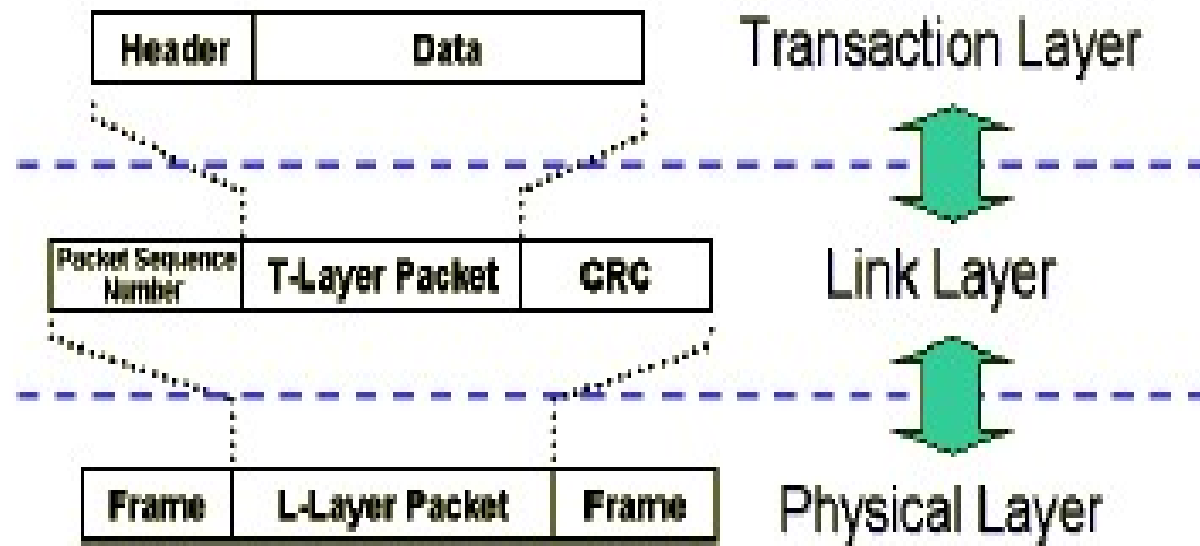


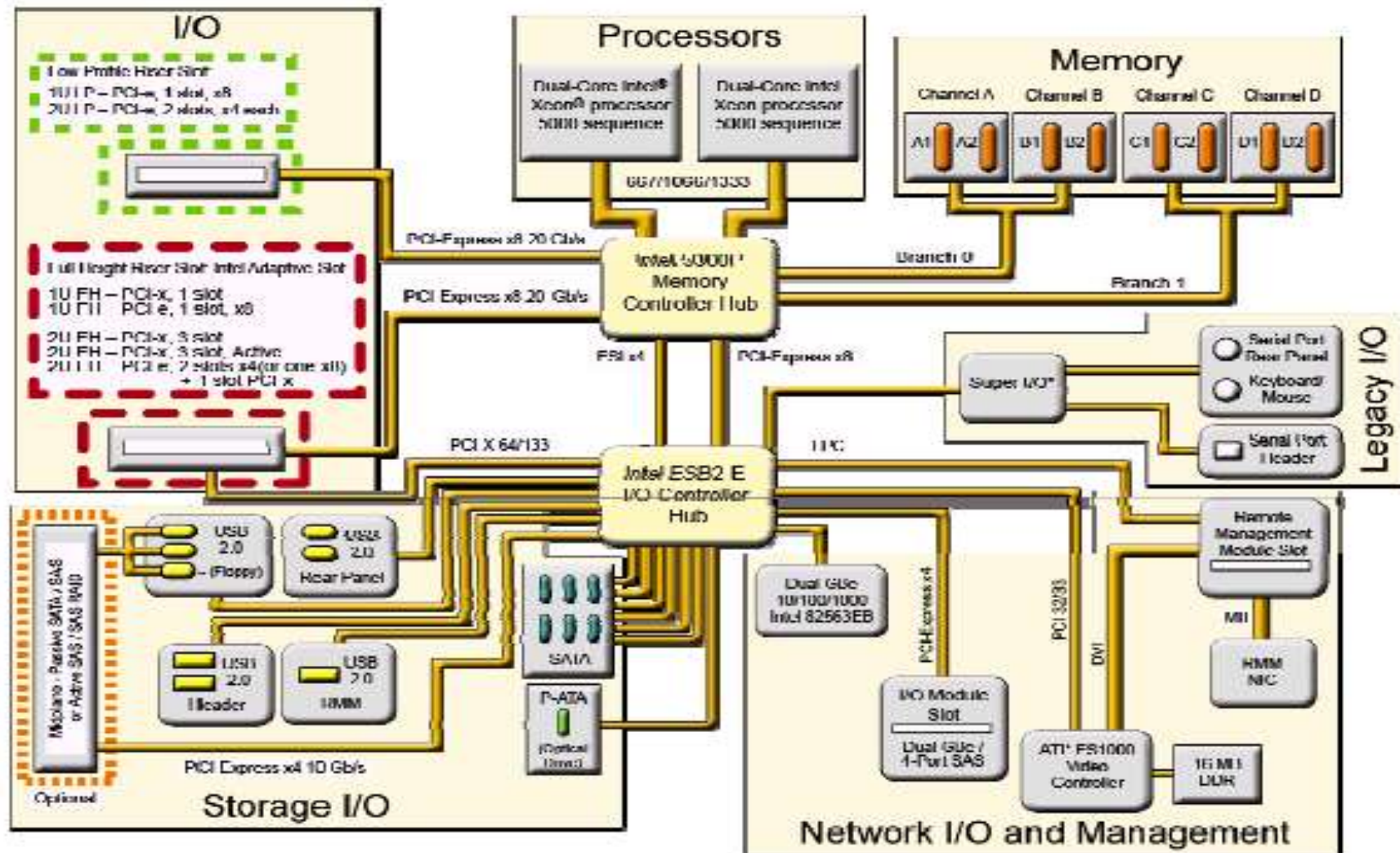
Figure 9. A PCI Express Link consists of one or more lanes

PCI Express Link Layer



The Link Layer adds data integrity features

INTEL S5000PAL



INTEL Core I7

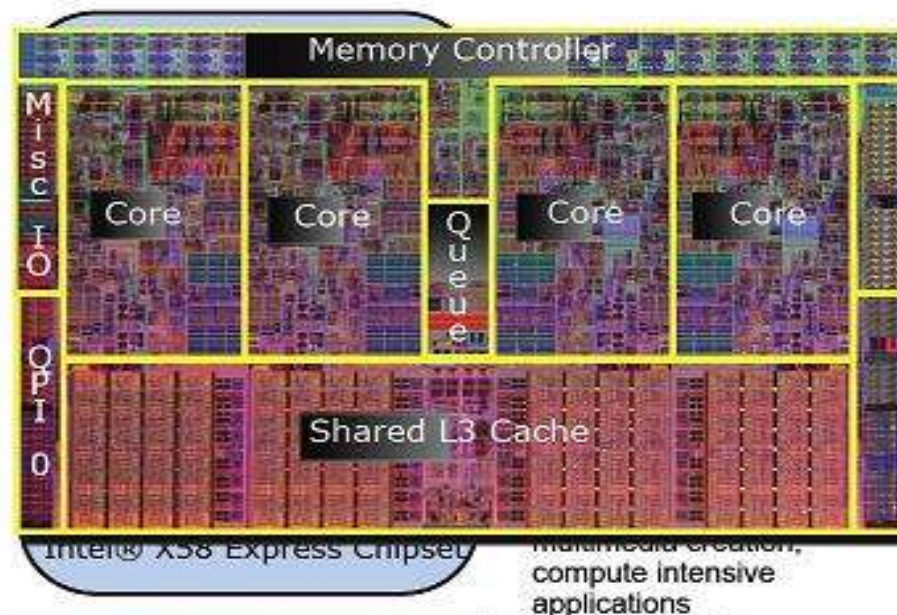
- QuickPath conexión de 25.6 GB/s a 20-bit. Esta conexión provee exactamente el doble del ancho de banda teórico de un FSB
 - la arquitectura QuickPath asume que la placa tiene un controlador de memoria integrado, obligando así a los multiprocesadores (multiplacas) a usar una arquitectura NUMA (verlo como una mesh luego siguen un esquema NUMA)
- Memoria de tres canales: cada canal puede soportar una o dos memorias DIMM DDR3 (en la mother deben ser colocadas en grupos de 3)
- Cada nucleo (core) es HT y tiene L2 de 256 KB
- Cache L3 de 8 MB de memoria, compartida por todos los núcleos
- 45 nm próximos en 32 nm (11/2014 14 nm)

INTEL Core I7

Intel® Core™ i7 Processor

Performance/Features:

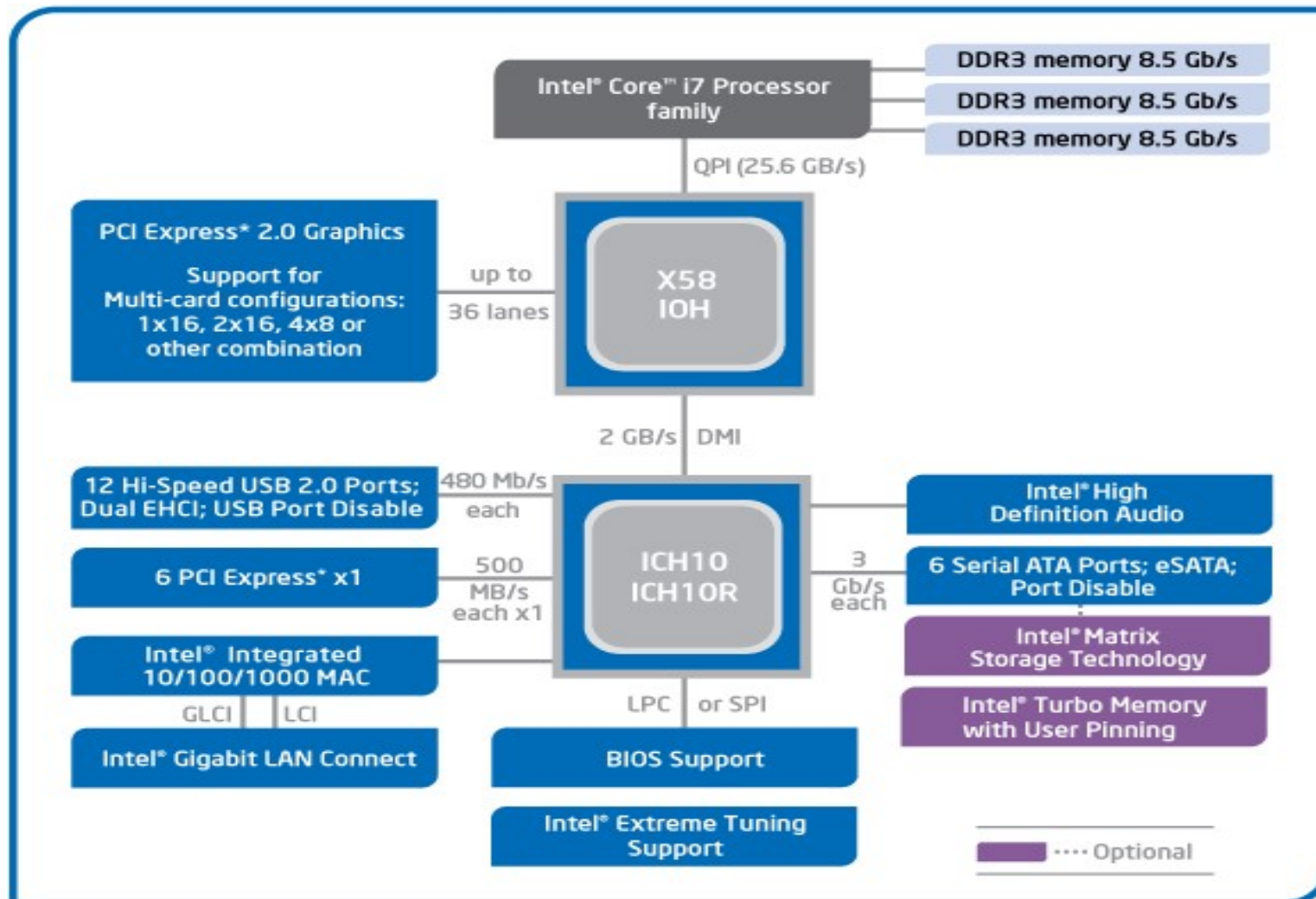
- 8 processing threads via Intel® Hyper-Threading Technology (HT)
- 4 cores
- Turbo Mode operation
- Intel® QuickPath Interconnect (Intel® QPI) to Intel® X58 Express Chipset
- Integrated Memory Controller (IMC) – 3ch DDR3
- 7 more SSE4 instructions
- Overspeed Protection Removed



Intel's Next Gen Computing Genius!

Intel Developer
DEMO FORUM

INTEL Core I7

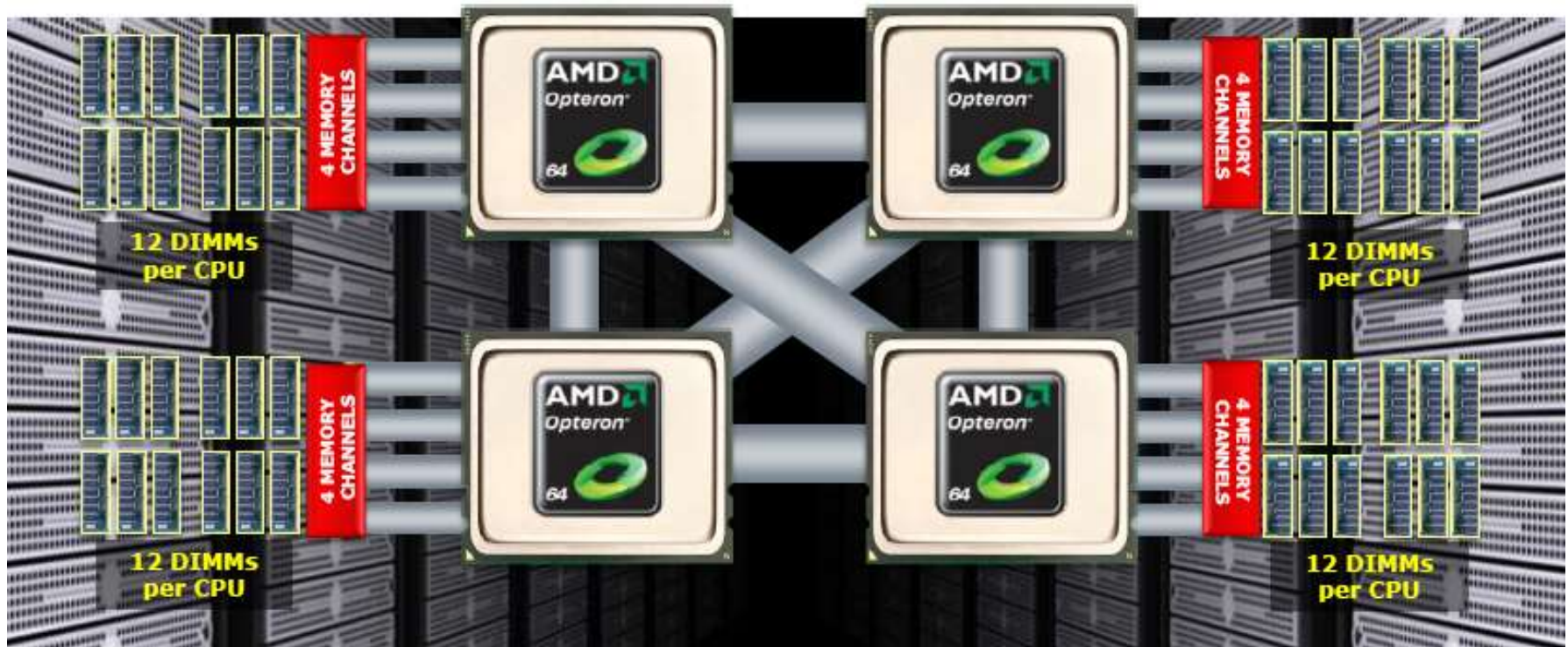


Intel® X58 Express Chipset Block Diagram

AMD OPTERON 6000

Direct Connect Architecture 2.0

Balanced and Scalable Design to Support up to 16 Cores per CPU²



AMD OPTERON 6000

AMD-P Technologies

AMD PowerCap Manager

Allows IT datacenter managers to set a fixed limit on a server's processor power consumption

AMD PowerNow!™ Technology with Independent Dynamic Core Technology

Allows processors and cores to dynamically operate at lower power and frequencies, depending on usage and workload to help reduce TCO and to lower power consumption in the datacenter

AMD Smart Fetch Technology

Can reduce power consumption by allowing idle cores to enter a "halt" state

AMD CoolCore™ Technology

Can reduce processor power consumption by dynamically turning off sections of the processor when inactive

AMD CoolSpeed Technology

Highly accurate thermal information & thermal protection

Advanced Processor Management Link*

Allows advanced power control and thermal policies

Low Power U/RDDR3 memory

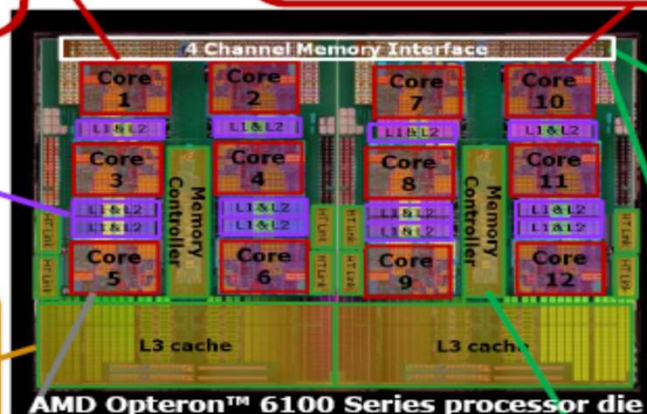
Supports DDR3 1.5v and low power DDR3L 1.35v memory technologies

Dual Dynamic Power Management

Enables more granular power management capabilities to reduce processor energy consumption. Separate power planes for cores and memory controller

C1E

Reduces memory controller and Hypertransport™ technology links' power



* In APML enabled systems

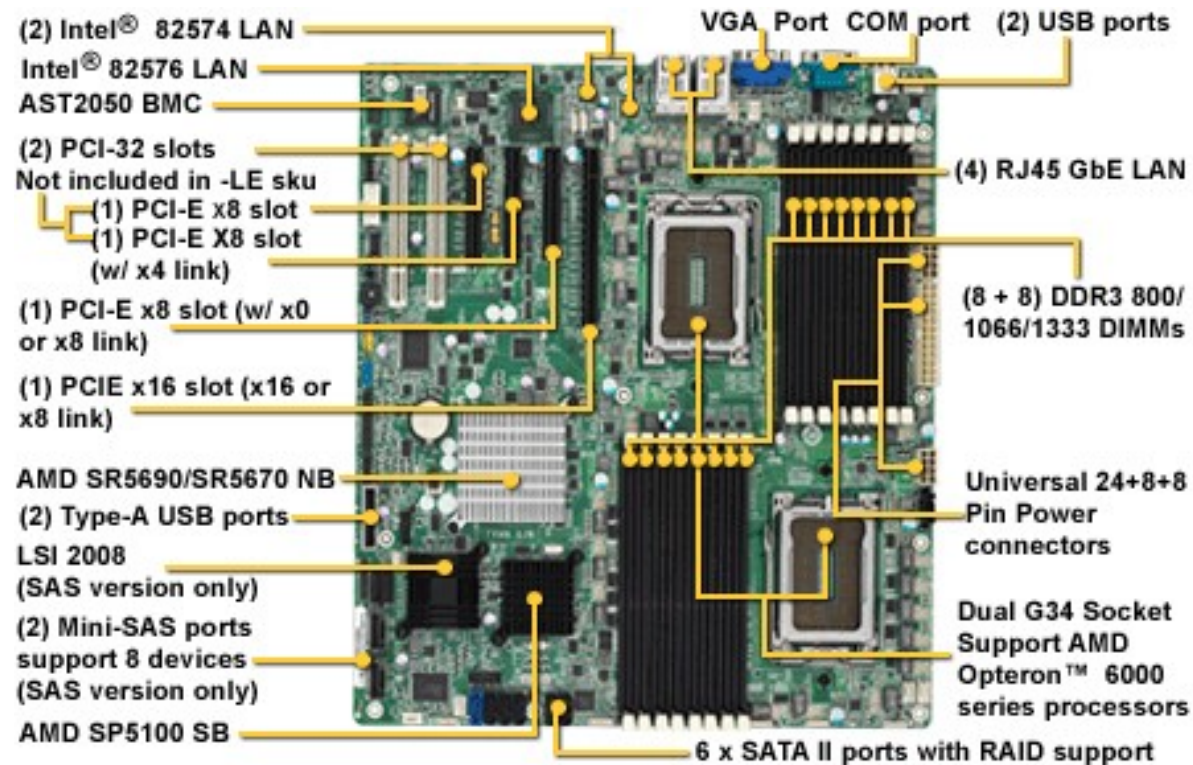


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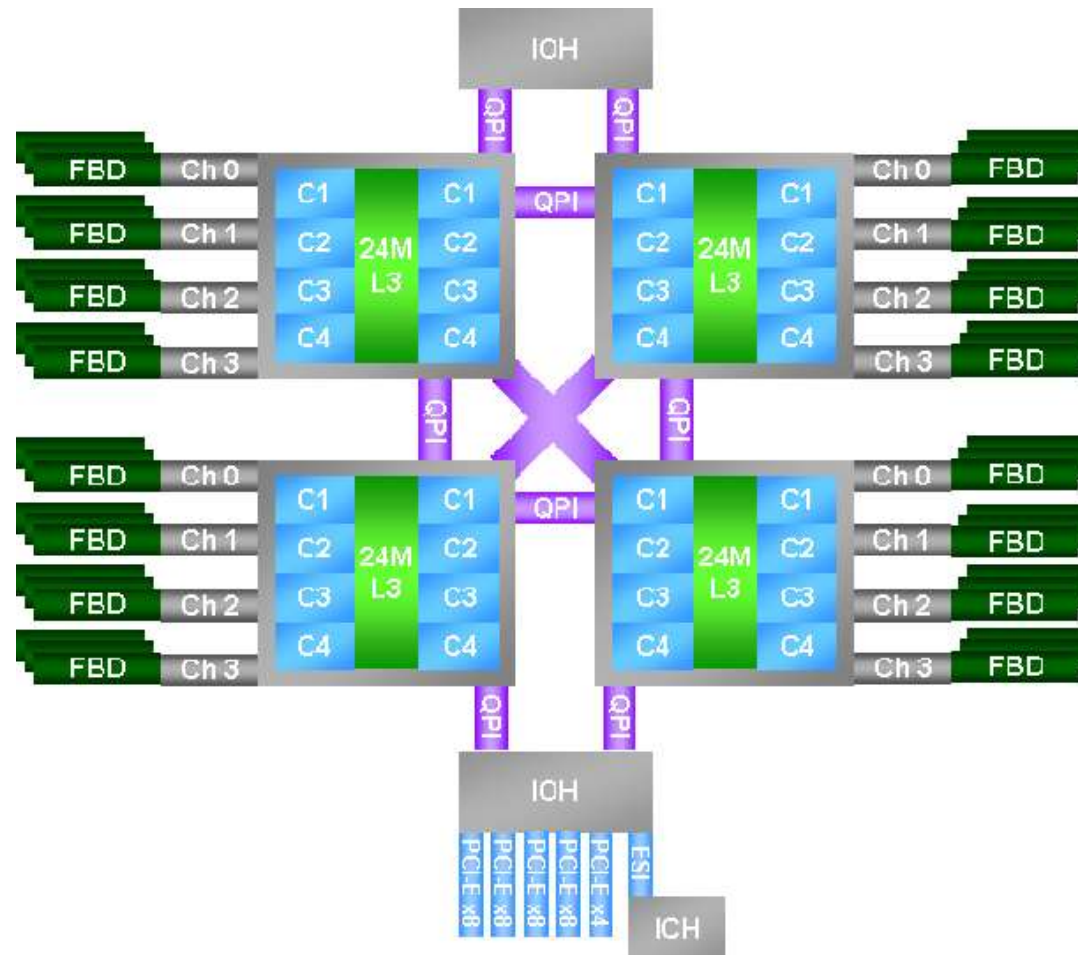
Server 1H 2010 Solution Provider Roadmap - AMD Confidential



MotherBoard Tyan



INTEL 5520 IOH



The diagram shows a 4-way Xeon 7500 system with two IOH devices

IOH High-Level Block Diagram

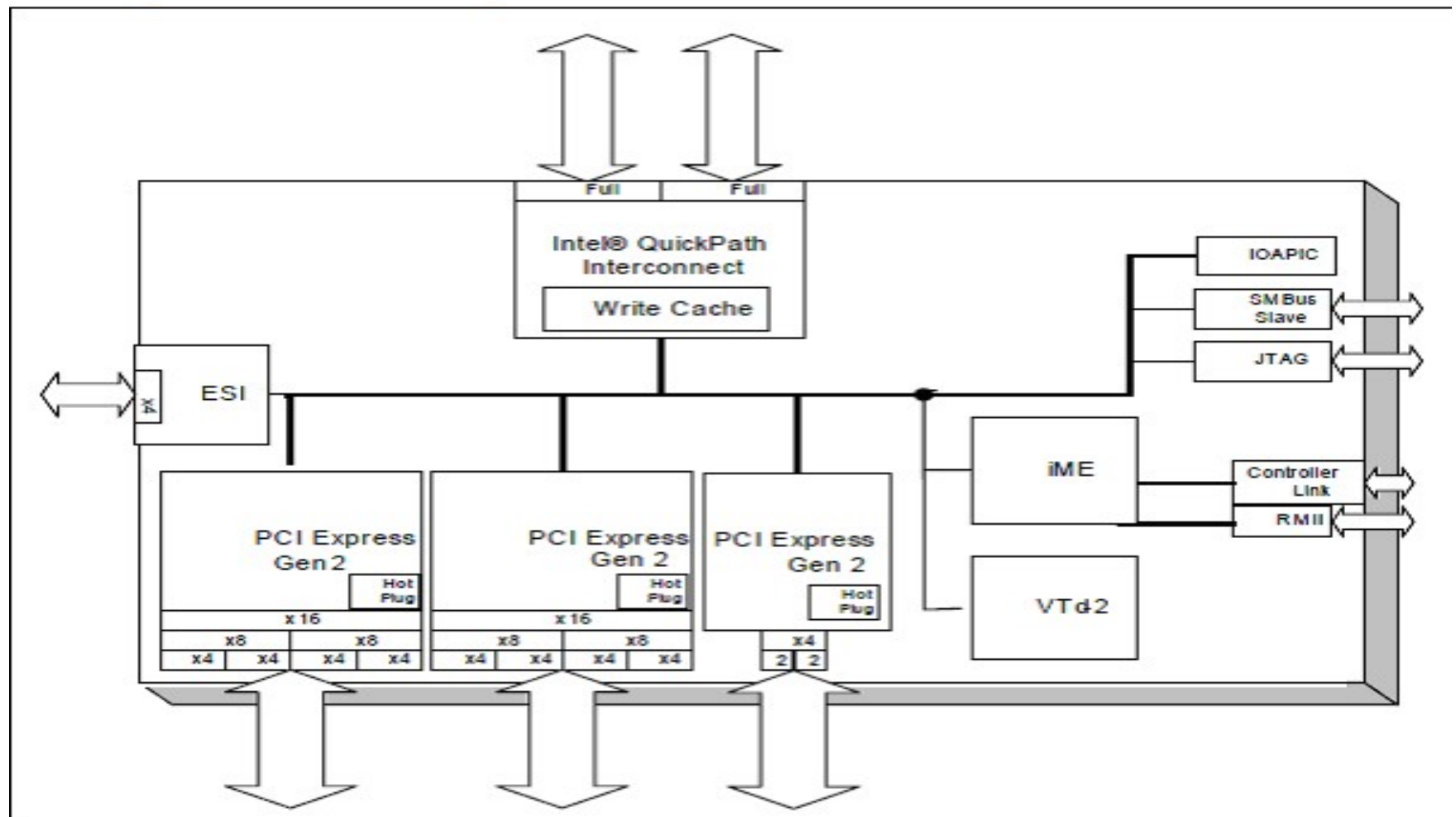
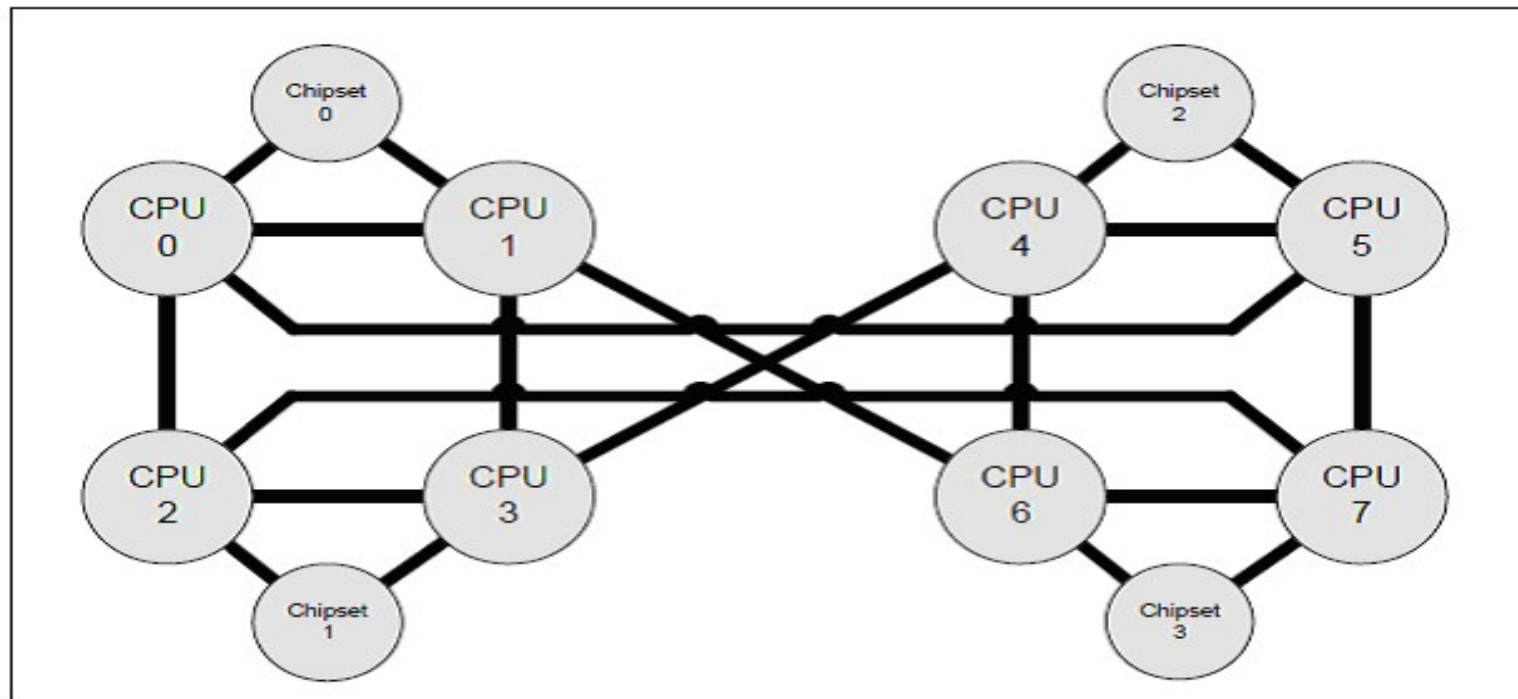


Diagrama de un INTEL Xeon octo 7500



GPUS en el Cálculo

Se comenzó con Procesadores gráficos de
cálculo entero. (multiprocesadores) y
derivó

en

Nvidia Tesla GPU

Con compiladores

CUDA opencl ...

Por comenzar en 2012/13

Xeon Phi (Knight's Corner)

(22 nm) (14nm (desde 11/2014))

deriva de Larrabee GPU

57, 60 or 61 cores en un solo módulo

run standard x86 code

1 TFLOPS DP (double precision)

2 TFLOPS SP (single-precision)

Más

1.8 MB to 1.9 MB of L1 cache

28 MB to 30.5 MB of L2 cache

3GB, 6GB or 8GB onboard GDDR5
memory

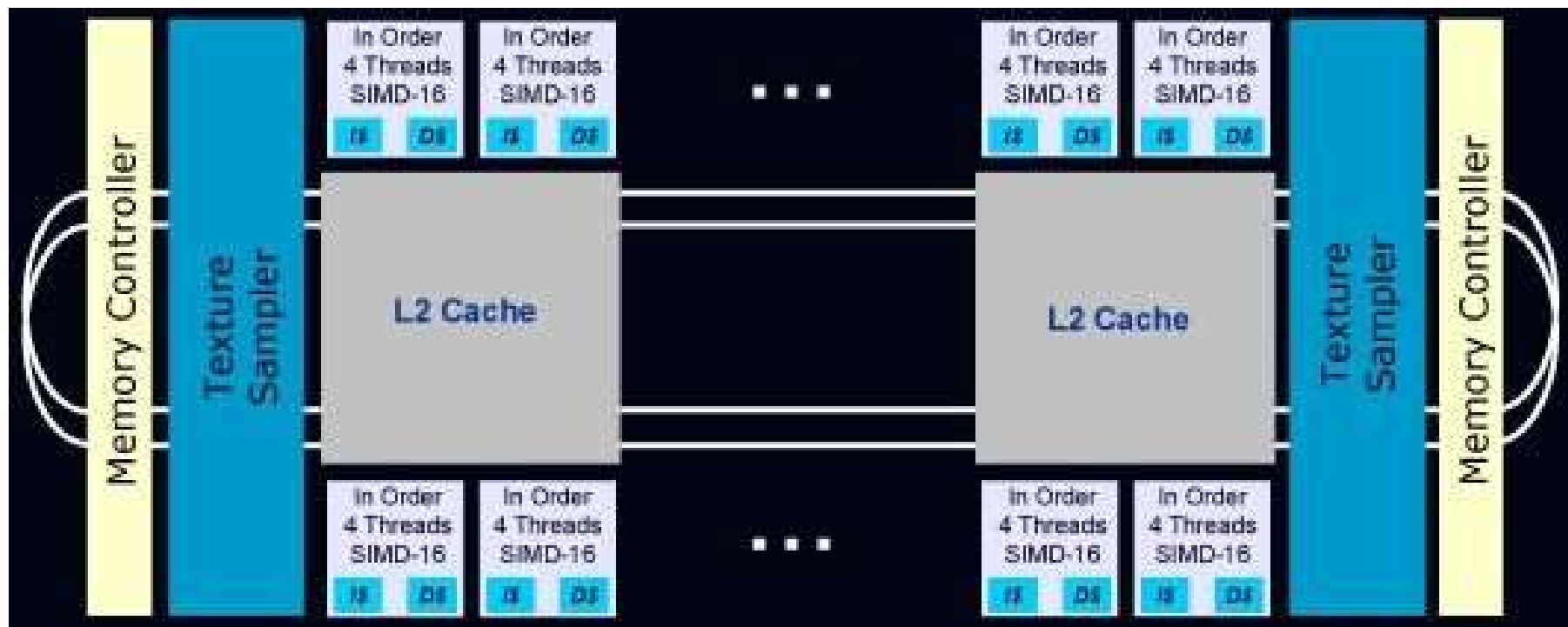
clock speed of 600 MHz –630 MH

57 core Xeon Phi models will have a
maximum clock of 1.1 GH

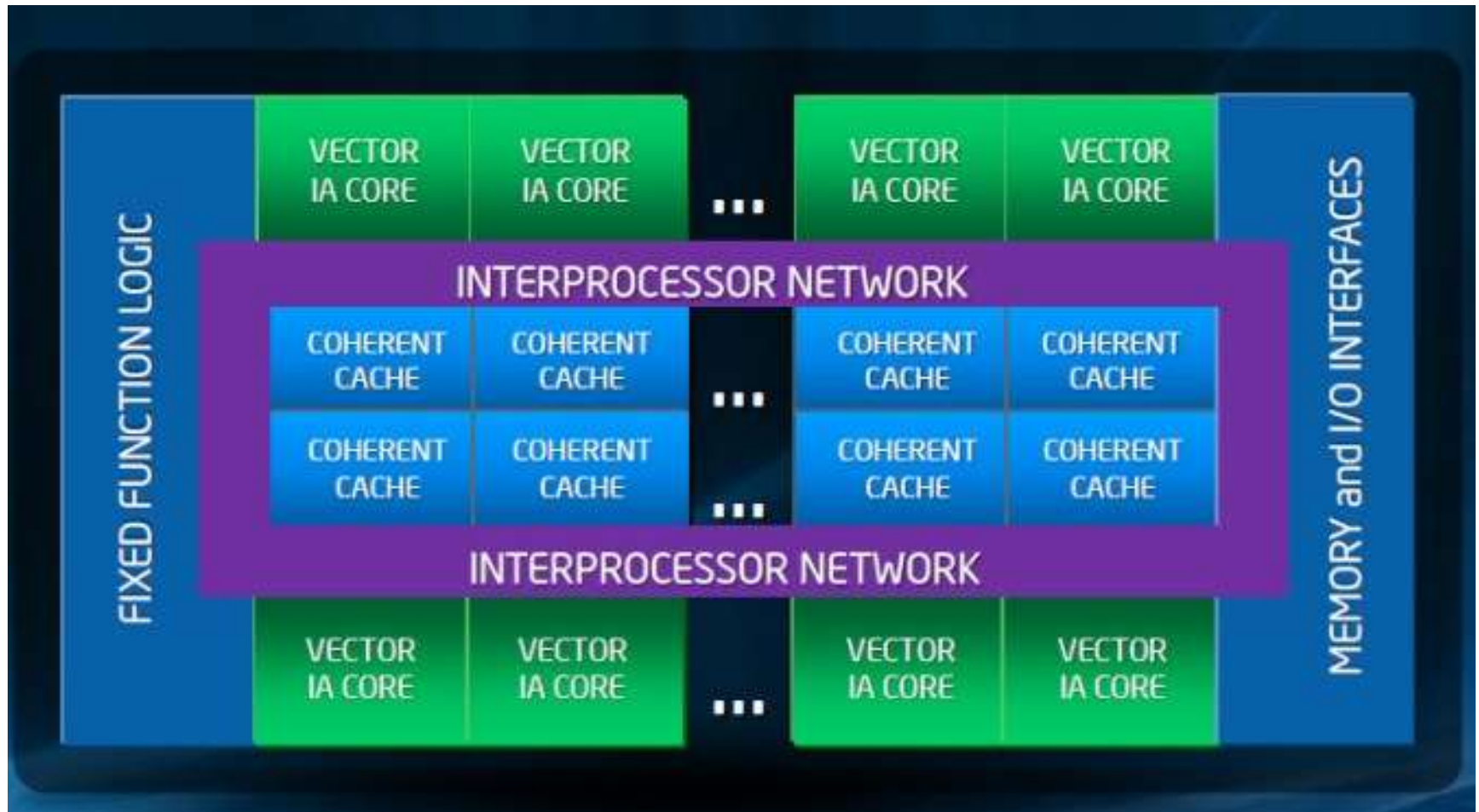
MIC

- . The memory will be clocked at 1.25 - 1.375 GHz giving in excess of 300 GBb/s bandwidth.
- . TDP of 245W -TDP of 300W
- . Cooling (passive /active)
- . KNC can communicate with other MIC cards across the PCIe bus
- . Convive con x86/x86-64 (Xeon E5)
- . Pueden utilizarse bibliotecas x86/x86-64

Larabee



Xeon Phi MIC block diagram



On the left is a Xeon Phi "core;" on the right is the vector unit. Source: Intel via ExtremeTech

